

1.3 Value Types

As mentioned above, each of these hives contain keys and sub-keys, which in turn can contain further sub-keys or values. These values are of three major types: DWORD, Binary and String.

REG_DWORD (a DWORD Value) is usually used for Boolean values and is a four-byte number. Many device drivers and services use DWORD values to toggle between options. For example, the UpdateMode setting that controls refresh rates can have a setting of either 0 (disabled) or 1 (enabled). If the DWORD value is set to 0, refreshing does not take place. Each setting has a specific default DWORD value that is used by the system.

REG_BINARY (a Binary Value) is used to store information as raw binary data, and is usually used for hardware components.

The String Value type, however, has expansions to accommodate variables and multiple values. The Value types are denoted as:

REG_SZ is the standard string used to display human-readable text.

REG_EXPAND_SZ is an expandable data string that permits storing of variables that can be replaced by actual values by the application calling the key. For example, an application may refer to a particular key in a hive to obtain the location of a system file. The key would contain a setting for that system file, and a string value which is, say, `%systemroot%\filename`. The `%systemroot%` will be replaced by the location of the XP operating system folder, which on most machines will be `C:\Windows` or `C:\WINNT`.

REG_MULTI_SZ is used to store lists or multiple values, each entry being separated by a NULL character. This is analogous to arrays in programming. For example, a Registry entry could be created to store the IP addresses of multiple timeservers. An application program would then refer to this Registry entry and cycle through the list of IP addresses.